Bramhall Copper Corp.

Quebrada Azul Copper Expansion

GRINDING MILL DATA SHEET

QAZ-EP-3100-255-DST-55017-Q

Revision Q | 2021-08-26 | Bid issue

Discipline 255, area 3100

Contractor: Ostrander Engineering Services Ltd.

Drive vendor: Veltrane Drives GmbH

Project number 7-0112, bid tag 7-0112-BQ

Mills: 3210-MLL-01 (SAG mill) and 3210-MLL-02 (ball mill)

Drives: 3210-MTR-01 (SAG) and 3210-MTR-02 (ball)

IDENTIFICATION

A. PROJECT AND CONTRACT	
Client	Bramhall Copper Corp.
Project	Quebrada Azul Copper Expansion
Contractor	Ostrander Engineering Services Ltd.
Project number	7-0112
Bid tag	7-0112-BQ
Region	Norte Alto Province
Nearest town	Calderón Verde
Distance to regional capital	620 km
Site altitude	3,850 masl
Drive vendor	Veltrane Drives GmbH
Local authority	Local Authority Having Jurisdiction (LAHJ), Norte Alto Province electrical code
Electrical code	Norte Alto Electrical Code NAEC-2019
Document number	QAZ-EP-3100-255-DST-55017-Q
Discipline and area	255 (area 3100)

B. MILL AND DRIVE EQUIPMENT

Tag	Equipment	Area	Rating	Impedance	Voltage	Poles	Diameter (m)	Eff. length (m)	Documents
3210-MLL-01	SAG mill	3210 drive area	—	—	—	—	9.8	5.1	GMDS, DST
3210-MLL-02	Ball mill	3210 drive area	—	—	—	—	7.9	11.8	GMDS, DST
3210-MTR-01	SAG gearless drive motor	3210 drive area	16,500 kW	—	34.5 kV	72	—	—	SPC, DST, GMDS, SLD
3210-MTR-02	Ball gearless drive motor	3210 drive area	12,000 kW	—	34.5 kV	64	—	—	SPC, DST, GMDS, SLD
3210-XFR-01	Excitation transformer	3210 drive area	1.6 MVA	6.0%	34.5 kV / 380 V	—	—	—	SPC, DST
3210-XFR-02	Cyclo-converter transformer, SAG	3210 drive area	20.0 MVA	7.5%	—	—	—	—	SPC, DST
3210-XFR-03	Cyclo-converter transformer, ball	3210 drive area	14.0 MVA	7.5%	—	—	—	—	SPC, DST
3210-LCP-01	Mill local control panel	3210 drive area	—	—	—	—	—	—	SPC, DST

C. RELATED DOCUMENTS

Document number	Title	Rev	Date
QAZ-EP-0000-265-SPC-00006-Q	GEARLESS MILL DRIVE SPECIFICATION (SAG MILLS AND BALL MILLS)	Q	2021-08-26
QAZ-EP-0000-265-DST-00006-Q	GEARLESS MILL DRIVE DATA SHEET	Q	2021-08-26
QAZ-EP-0000-210-DBD-00002-B	GENERAL SITE CONDITIONS	B	2019-10-02
QAZ-5200-265-E-001_B	SINGLE LINE DIAGRAM - MILL DRIVE POWER SUPPLY	B	2019-10-02

D. GOVERNING STANDARDS

Motors	IEC 60034-1	Motors (NEMA)	NEMA MG 1
Harmonics	IEEE 519-2014	Transformers	ANSI C57.12.00
Quality	ISO 9001:2015	Electrical code	NAEC-2019 and LAHJ

E. SUPPLY BASIS

MV bus	23/34.5 kV	Drive supply	34.5 kV
Auxiliary bus	6.6 kV	Low voltage bus	380 V
Status	Issued for bid under revision Q, 2021-08-26		

REVISION AND APPROVAL

Rev	Date	Description	Originator	Checker	Approver	Doc. controller
A	2019-09-06	For info	Superseded by revision Q			
B	2019-10-02	Comments incorporated	Superseded by revision Q			
Q	2021-08-26	Bid issue	Tamsin Quell	Renata Okafor	Simon Veil	Paulo Maren

MILL MECHANICAL DATA: 3210-MLL-01 SAG MILL

No.	Parameter	Symbol	Unit	Value	Basis / calculation	Source
1	Mill tag			3210-MLL-01	Tag per plant list	GMDS
2	Mill type			SAG mill	Mill service	SPC
3	Mill inside diameter	D	m	9.8	Inside diameter of shell	DST
4	Effective grinding length	L_eff	m	5.1	Between effective heads	DST
5	Critical speed constant			42.3	Constant in n_c = 42.3 / sqrt(D_eff)	DST
6	Critical speed	n_c	rpm	13.51	42.3 / sqrt(9.8) = 13.5122	DST
7	Operating speed	n_op	rpm	10.13	120 x 6.08 / 72 = 10.1333	DST
8	Operating speed as % of critical		%	75.0%	10.1333 / 13.5122 = 74.99%, shown as 75.0%	DST
9	Mill operating mode			Vendor to state	Process design basis	SPC
10	Drive rated power	P	kW	16,500	Rated mill drive power	SPC
11	Drive pole count	p		72	Gearless motor 3210-MTR-01	DST
12	Drive frequency at operating speed	f	Hz	6.08	Supply frequency	DST
13	Drive angular speed	omega	rad/s	1.061	2 x PI x 10.1333 / 60 = 1.06116	DST
14	Drive torque at operating speed	T	kN·m	15,549	16,500 / 1.06116 = 15,549.0	DST
15	Drive efficiency	eta		0.97	Electrical data basis	DST
16	Design power factor	pf		0.95	Electrical data basis	DST
17	Full-load current at 34.5 kV	I_FL	A	299.6	16,500 / (sqrt(3) x 34.5 x 0.95 x 0.97) = 299.646	DST
18	Drive supply voltage	V	kV	34.5	Medium voltage drive supply	SLD
19	Site altitude		masl	3,850	Site basis	SITE
20	Altitude derating factor			0.94	1 - 0.0000210 x (3,850 - 1,000) = 0.94015, rounded	SPC
21	Drive nameplate power		kW	17,553	16,500 / 0.94 = 17,553.2	DST
22	Input apparent power	S	kVA	17,906	16,500 / (0.95 x 0.97) = 17,905.6	DST
23	Cyclo-converter transformer			3210-XFR-02	SAG drive transformer	SPC
24	Transformer rating		MVA	20.0	Rated power of 3210-XFR-02	DST
25	Transformer impedance		%	7.5%	Impedance of 3210-XFR-02	DST
26	Transformer tap range			±2 x 2.5%	5 positions	DST
27	Mill local control panel			3210-LCP-01	Mill operator interface	SPC
28	Shell plate thickness		mm	Vendor to state	Vendor submittal	Vendor
29	Head plate thickness		mm	Vendor to state	Vendor submittal	Vendor
30	Empty mill mass		t	Vendor to state	Vendor submittal	Vendor
31	Charge mass at design load		t	Vendor to state	Vendor submittal	Vendor
32	Trunnion journal diameter		mm	Vendor to state	Vendor submittal	Vendor
33	Trunnion bearing type			Vendor to state	Vendor submittal	Vendor
34	Trunnion bearing lubricant			Vendor to state	Vendor submittal	Vendor
35	Lubricant flow per bearing		L/min	Vendor to state	Vendor submittal	Vendor
36	Mill rotation direction			Vendor to state	Relative to feed end	Vendor
37	Feed opening diameter		mm	Vendor to state	Vendor submittal	Vendor
38	Discharge opening		mm	Vendor to state	Vendor submittal	Vendor
39	Shell liner type			Vendor to state	Vendor submittal	Vendor
40	Head liner type			Vendor to state	Vendor submittal	Vendor
41	Lifter type			Vendor to state	Vendor submittal	Vendor
42	Lifter count per row			Vendor to state	Vendor submittal	Vendor
43	Liner material			Vendor to state	Vendor submittal	Vendor
44	Bearing temperature alarm		°C	Vendor to state	Vendor submittal	Vendor
45	Mill vibration alarm		mm/s	Vendor to state	Vendor submittal	Vendor

3210-MLL-01 SAG MILL: SHELL AND HEADS

Shell, heads, welding, inspection, openings, flanges, painting and mass data for SAG mill 3210-MLL-01.

This page covers the shell and heads of SAG mill 3210-MLL-01: 9.8 m inside diameter, 5.1 m effective grinding length, plus welding, inspection, openings, flanges and painting.

The shell transmits torque from drive 3210-MTR-01, 16,500 kW, at 10.13 rpm, 75.0% of the 13.51 rpm critical speed. Plate thickness and masses: Vendor to state.

No.	Parameter	Unit	Value	Basis	Remark
1	GENERAL				
2	Tag		3210-MLL-01		
3	Inside diameter	m	9.8		
4	Effective length	m	5.1		
5	Critical speed	rpm	13.51		
6	Operating speed	rpm	10.13	75.0%	
7	Drive		3210-MTR-01		
8	Drive power	kW	16,500		
9	SITE				
10	Min ambient	°C	-12		
11	Max humidity		95%		
12	SHELL				
13	Joints		Welded		
14	Material		Steel		
15	Plates		Rolled		
16	Liners		Bolted		
17	HEADS				
18	Type		Fabricated		
19	Material		Steel		
20	Trunnion		Integral		
21	Machining		Required		
22	WELDING				
23	Seams		Butt		
24	Procedure		Qualified		
25	Welders		Certified		
26	PWHT		Required		
27	INSPECTION				
28	Radiography		Yes		
29	Ultrasonic		Yes		
30	MPI		Yes		
31	Visual		Yes		
32	Records		Required		
33	OPENINGS				
34	Manhole		Yes		
35	Covers		Bolted		
36	Reinforcement		Required		
37	FLANGES				
38	Joint		Bolted		
39	Faces		Machined		
40	Bolting		Alloy		
41	PAINTING				
42	Primer		Epoxy		
43	Finish		Polyurethane		
44	Preparation		Blasted		
45	MASS				
46	Weighing		Required		
47	Balance		Static		
48	Lifting		Lugs		
49	Drawing		Vendor		
50	Certificate		Required		

3210-MLL-01 SAG MILL: TRUNNIONS AND BEARINGS

SAG mill 3210-MLL-01 trunnion journals, bearings, seals, pedestals and condition monitoring requirements.

Trunnion journals on 3210-MLL-01 carry the 9.8 m by 5.1 m shell on hydrostatic bearings set in fabricated housings and pedestals. Labyrinth seals keep grinding slurry out of the oil.

At 10.13 rpm, 75.0% of critical, the 16,500 kW drive 3210-MTR-01 is interlocked with bearing temperature and vibration detectors. Rating uses the 0.94 derating at 3,850 masl.

No.	Parameter	Unit	Value	Basis	Remark
1	Mill				
2	Tag		3210-MLL-01		
3	Diameter	m	9.8		
4	Length	m	5.1		
5	Critical	rpm	13.51		
6	Operating	rpm	10.13		
7	Fraction		75.0%		
8	Drive		3210-MTR-01		
9	Power	kW	16,500		
10	Site				
11	Altitude	masl	3,850		
12	Derating		0.94		
13	Ambient	°C	30		
14	Minimum	°C	-12		
15	Maximum	°C	27		
16	Humidity		95%		
17	Zone		4	Seismic	
18	PGA	g	0.42		
19	Class		D		
20	Frost	mm	450		
21	Wind	m/s	36		
22	Trunnions				
23	Journal		Steel		
24	Finish		Polished		
25	Liner		Replaceable		
26	Flange		Bolted		
27	Inspection		Required		
28	Oil		Circulating		
29	Bearings				
30	Type		Hydrostatic		
31	Pads		Self-aligning		
32	Lining		Babbitt		
33	Housing		Fabricated		
34	Split		Horizontal		
35	Lift		High-pressure		
36	Fixed		Feed-end		
37	Free		Discharge-end		
38	Seals				
39	Type		Labyrinth		
40	Purge		Grease		
41	Guard		Yes		
42	Pedestal		Fabricated		
43	Grouting		Epoxy		
44	Monitoring				
45	RTD		Duplex		
46	Vibration		Accelerometer		
47	Alarm		Yes		
48	Trip		Yes		
49	Interlock		3210-MTR-01		
50	Oil-flow switch		Yes		

3210-MLL-01 SAG MILL: FEED AND DISCHARGE

Feed and discharge arrangement data for SAG mill 3210-MLL-01, covering chute, trunnions, grate, trommel and spillage.

The feed chute delivers ore and water into the feed trunnion of SAG mill 3210-MLL-01, which has a 9.8 m inside diameter and 5.1 m effective grinding length.

At the discharge end, the grate and pulp lifters pass slurry to the trommel. Running at 10.13 rpm, 75.0% of critical, under drive 3210-MTR-01 of 16,500 kW, the lifters must clear pulp without backflow.

No.	Parameter	Unit	Value	Basis	Remark
1	Tag		3210-MLL-01		
2	Diameter	m	9.8		
3	Length	m	5.1		
4	Critical	rpm	13.51	Formula	
5	Speed	rpm	10.13		
6	Fraction		75.0%		Of critical
7	Drive		3210-MTR-01		
8	Power	kW	16,500		Main drive
9	FEED-CHUTE				
10	Construction		Welded		
11	Lining		Replaceable		
12	Mounting		Retractable		
13	Sealing		Yes		
14	Inspection		Hatch		
15	Feed-spout		Removable		
16	FEED-TRUNNION				
17	Liner		Bolted		
18	Bearing		Hydrostatic		
19	Lubrication		Yes		
20	Spiral		No		
21	Access		Manhole		
22	DISCHARGE-TRUNNION				
23	Liner		Replaceable		
24	Bearing		Hydrostatic		
25	Seal		Labyrinth		
26	GRATE				
27	Segments		Bolted		
28	Material		Alloy-steel		
29	Apertures		Slotted		
30	Wear		Monitored		
31	PULP-LIFTERS				
32	Type		Radial		
33	Material		Rubber		
34	Backflow		Minimised		
35	TROMMEL				
36	Mounting		Flanged		
37	Screen		Polyurethane		
38	Spray		Yes		
39	Oversize		Recycled		
40	Undersize		Sump		
41	Guard		Enclosed		
42	SPILLAGE				
43	Bunding		Yes		
44	Floor		Sloped		
45	Sump		Yes		
46	Pump		Submersible		
47	Washdown		Yes		
48	Destination		Sump		
49	Access		Vehicle		
50	Drainage		Trenched		

3210-MLL-01 SAG MILL: SHELL LINERS

Shell liner types, materials, fixing, handling, spares and wear notes for SAG mill 3210-MLL-01.

The shell liners protect the 9.8 m diameter shell of SAG mill 3210-MLL-01 over the 5.1 m grinding length and lift the charge as the mill turns.

Lifter profile suits operation at 10.13 rpm, which is 75.0% of the 13.51 rpm critical speed, with drive 3210-MTR-01 rated 16,500 kW. Wear allowance is Vendor to state.

No.	Item	Location	Type	Material	Fixing	Spare	Value/Note
1	Tag 3210-MLL-01						
2	Diameter						9.8 m
3	Length						5.1 m
4	Speed						10.13 rpm
5	Critical speed						13.51 rpm
6	Speed ratio						75.0%
7	Drive 3210-MTR-01						16,500 kW
8	Feedhead		Plate				
9	Grates		Slotted				
10	Lifters		Bar				
11	Plates	Shell					
12	Lifters			CrMo			
13	Plates			CrMo			
14	Grates			CrMo			
15	Fixing				Bolted		
16	Bolts		Wedge				
17	Seals						Required
18	Backing			Rubber			
19	Handler		Remote				
20	Lugs						Required
21	Spares					Full	
22	Bolts					Yes	
23	Lifters					Yes	
24	Plates					Yes	
25	Grates					Yes	
26	Wear						Profiled
27	Pulp lifters			CrMo			
28	Cone	Discharge					
29	Trunnion			Rubber			
30	Fillers				Bolted		
31	Shims			Steel			
32	Gaskets			Rubber			
33	Torque						Tool
34	Inching						Required
35	Lock						Required
36	Storage						Indoor
37	Crates						Steel
38	Marking						Cast
39	Drawings						Required
40	Inspection						Shop
41	Hardness						Tested
42	Fit-up						Trial
43	Orientation						Marked
44	Faces		Angled				
45	DEM						Required
46	Relining						Planned
47	Access	Feed					
48	Holes						Tapered
49	Sensors						Optional
50	Certificates						Required

3210-MLL-01 SAG MILL: LIFTER SCHEDULE

Lifter schedule for SAG mill 3210-MLL-01 covering rows, profile, base, bolting, material and replacement set.

This page lists the lifters for SAG mill 3210-MLL-01, which has inside diameter of 9.8 m and an effective grinding length of 5.1 m. Row count, profile height and set mass are for the vendor to confirm.

Lifter geometry must suit the operating speed of 10.13 rpm, 75.0% of the 13.51 rpm critical speed, and the charge motion from drive 3210-MTR-01 at 16,500 kW.

No.	Value / Entry	Item	Unit	Basis	Source	Status	Ref	Remark
1	MILL							
2	3210-MLL-01	Tag						
3	9.8	ID	m					
4	5.1	EGL	m					
5	13.51	Nc	rpm					
6	10.13	Speed	rpm					
7	75.0%	Fraction						
8	3210-MTR-01	Drive						
9	16,500	Power	kW					
10	LIFTERS							
11	Vendor to state	Rows						
12	Hi-Lo	Pattern						
13	Vendor to state	Height						
14	Trapezoid	Profile						
15	Integral	Base						
16	BOLTING							
17	Through	Type						
18	Metric	Thread						
19	Rubber	Seal						
20	Captive	Nut						
21	MATERIAL							
22	Cr-Mo	Lifter						
23	Alloy	Bolts						
24	SPARES							
25	Full	Set						
26	Vendor to state	Mass						
27	Yes	Handler						
28	NOTES							
29	OEM	Drawing						

3210-MLL-01 SAG MILL: HEAD AND DISCHARGE LINERS

SAG mill 3210-MLL-01 feed head liners, discharge grates, pulp lifters and sealing data

This page lists head and discharge liner data for SAG mill 3210-MLL-01, which has an inside diameter of 9.8 m and an effective grinding length of 5.1 m. Feed head liners, grates and pulp lifters protect the heads and control discharge.

The mill runs at 10.13 rpm, 75.0% of the 13.51 rpm critical speed, driven by 3210-MTR-01 at 16,500 kW. Lifters suit this speed.

No.	Parameter	Unit	Value	Remark
1	Tag		3210-MLL-01	
2	Diameter	m	9.8	
3	Length	m	5.1	
4	Critical-speed	rpm	13.51	
5	Speed	rpm	10.13	75.0%
6	Drive	kW	16,500	3210-MTR-01
7	Feed-head			
8	Liners		Cr-Mo	
9	Lifters		Integral	
10	Fixing		Bolted	
11	Discharge-head			
12	Liners		Cr-Mo	
13	Grates		Cr-Mo	
14	Apertures		Tapered	
15	Open-area	%	Vendor to state	
16	Pulp-lifters		Radial	
17	Lifter-liners		Rubber	
18	Cone-liners		Cr-Mo	
19	Sealing			
20	Bolt-seals		Rubber	
21	Feed-seal		Lip	
22	Discharge-seal		Lip	
23	Gaskets		Yes	
24	Backing		Rubber	
25	Handling			
26	Lifting-lugs		Yes	
27	Liner-handler		Required	
28	Wear-gauging		Yes	
29	Spares		Recommended	
30	Reversible		Yes	

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Q	2021-08-26	Bid issue	Tamsin Quell	Renata Okafor	Simon Veil	Paulo Maren

NOTES TO SAG MILL MECHANICAL DATA

These notes record the basis for the SAG mill mechanical data. Mill 3210-MLL-01 has a 9.8 m inside diameter and 5.1 m effective grinding length, running at 10.13 rpm, or 75.0% of its 13.51 rpm critical speed.

Drive 3210-MTR-01 is rated 16,500 kW, derated by 0.94 for 3,850 masl. Unlisted values remain vendor responsibility.

No.	Note	Applies to
1	Inside diameter 9.8 m	3210-MLL-01
2	Effective length 5.1 m	
3	Critical speed 13.51 rpm	
4	Operating speed 10.13 rpm	
5	Speed 75.0% of critical	
6	Drive rating 16,500 kW	3210-MTR-01
7	Derating factor 0.94	
8	Site altitude 3,850 masl	All equipment
9	Design ambient 30 °C	
10	Extreme minimum -12 °C	
11	Extreme maximum 27 °C	
12	Design wind 36 m/s	Structures
13	Seismic zone 4	Foundations
14	PGA 0.42 g, class D	
15	Frost depth 450 mm	
16	Maximum humidity 95%	Electrical
17	Liner mass: Vendor to state	Liners
18	Shell thickness: Vendor to state	Shell
19	Vendor confirms mechanical values	
20	Speeds based on effective diameter	3210-MLL-01
21	Not for construction	

3210-MLL-01 SAG MILL: DRIVE MOTOR INTERFACE

Gearless motor to SAG mill 3210-MLL-01 interface: rotor mounting, air gap, bolting, alignment and access data.

The rotor of gearless motor 3210-MTR-01 is mounted directly on the shell flange of SAG mill 3210-MLL-01, which has an inside diameter of 9.8 m. The stator surrounds the rotor with a uniform air gap set by the vendor.

At 10.13 rpm, 75.0% of critical, the 16,500 kW drive delivers 15,549 kN·m, so bolting and alignment must hold the gap under load.

No.	Parameter	Value	Unit	Remark
1	Mill			
2	Tag	3210-MLL-01		
3	Diameter	9.8	m	Inside
4	Length	5.1	m	
5	Critical	13.51	rpm	
6	Speed	10.13	rpm	
7	Ratio	75.0%		
8	Motor			
9	Tag	3210-MTR-01		
10	Rating	16,500	kW	
11	Efficiency	0.97		
12	Voltage	34.5	kV	
13	Poles	72		
14	Frequency	6.08	Hz	
15	Torque	15,549	kN·m	
16	Derating	0.94		
17	Interface			
18	Rotor	Shell-mounted		
19	Attachment	Flange		
20	Airgap	Vendor to state		
21	Gap-sensors	Yes		
22	Stator	Split		
23	Stator-base	Grouted		
24	Rotor-segments	Bolted		
25	Bolting	Prestressed		
26	Bolt-tensioning	Hydraulic		
27	Locking	Yes		
28	Shell-flange	Machined		
29	Alignment	Laser		
30	Alignment-check	Periodic		
31	Thermal-growth	Allowed		
32	Enclosure	Sealed		
33	Access	Platforms		
34	Inspection	Hatches		
35	Lifting	Crane		
36	Inching	Yes		
37	Lock-out	Required		
38	Grounding	Shell		

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B	2019-10-02	Comments incorporated	Superseded by revision Q			
Q	2021-08-26	Bid issue	Tamsin Quell	Renata Okafor	Simon Veil	Paulo Maren
Status		Issued for bid, revision Q				

3210-MTR-01 SAG DRIVE: ELECTRICAL DATA

Electrical ratings, site conditions and control features of SAG gearless drive motor 3210-MTR-01.

Gearless motor 3210-MTR-01 drives SAG mill 3210-MLL-01 at 10.13 rpm, which is 75.0% of critical speed. The 72-pole motor runs at 6.08 Hz and delivers 15,549 kN·m at 16,500 kW shaft power.

At 3,850 masl an altitude derating factor of 0.94 raises the nameplate rating to 17,553 kW. The 34.5 kV supply carries 299.6 A and 17,906 kVA.

No.	Parameter	Unit	Value	Basis	Source	Remark
1	MOTOR					
2	Tag		3210-MTR-01			
3	Type		Gearless			
4	Mounting		Ring			
5	Rotor		Salient			
6	Stator		Segmented			
7	Cooling		Air			
8	Enclosure		Enclosed			
9	Insulation		Epoxy			
10	RATINGS					
11	Power	kW	16,500			
12	Nameplate	kW	17,553			
13	Voltage	kV	34.5			
14	Current	A	299.6			
15	Apparent	kVA	17,906			
16	PF		0.95			
17	Efficiency		0.97			
18	Poles		72			
19	Frequency	Hz	6.08			
20	Speed	rpm	10.13			
21	Angular	rad/s	1.061			
22	Torque	kN·m	15,549			
23	Derating		0.94			
24	MILL					
25	Mill		3210-MLL-01			
26	Diameter	m	9.8			
27	Length	m	5.1			
28	Critical	rpm	13.51			
29	Fraction		75.0%			
30	SITE					
31	Altitude	masl	3,850			
32	Ambient	°C	30			
33	Minimum	°C	-12			
34	Maximum	°C	27			
35	RH-mean		58%			
36	RH-max		95%			
37	CONTROL					
38	Converter		Cyclo			
39	Excitation		Static			
40	Inching		Yes			
41	Creep		Yes			
42	Braking		Yes			
43	Lock-detect		Yes			
44	PROTECTION					
45	Earthing		Resistance			
46	Harmonics		Filtered			
47	Airgap		Monitored			
48	AUXILIARIES					
49	Heaters		Yes			
50	Vibration		Monitored			

3210-MTR-01 SAG DRIVE: TRANSFORMERS AND EXCITATION

Transformer, rectifier and protection data for the 3210-MTR-01 SAG gearless drive supply

Cyclo-converter transformer 3210-XFR-02 is rated 20.0 MVA with 7.5% impedance and 5 tap positions. It supplies the 16,500 kW, 34.5 kV gearless motor 3210-MTR-01, whose input demand of 17,906 kVA sits within the rating.

Excitation transformer 3210-XFR-01, rated 1.6 MVA at 34.5 kV / 380 V with 6.0% impedance, feeds the field rectifier. Cooling and protection details rest with the vendor.

No.	Parameter	Unit	Value	Basis	Remark
1	3210-XFR-02				
2	Duty		Cycloconverter		
3	Rating	MVA	20.0		
4	Impedance		7.5%		
5	Taps		±2x2.5%		
6	Positions		5		
7	Load	kVA	17,906	Motor	
8	Phase shift		Multi-pulse		
9	Cooling		Oil-immersed		
10	Shield		Electrostatic		
11	Harmonics		Rated		
12	3210-XFR-01				
13	Duty		Excitation		
14	Rating	MVA	1.6		
15	Primary	kV	34.5		
16	Secondary	V	380		
17	Impedance		6.0%		
18	Cooling		Dry-type		
19	Rectifier				
20	Type		Thyristor		
21	Supply		3210-XFR-01		
22	Control		Digital		
23	Cooling		Air		
24	Redundant		Yes		
25	3210-MTR-01				
26	Power	kW	16,500		
27	Voltage	kV	34.5		
28	Current	A	299.6		
29	Frequency	Hz	6.08		
30	Protection provided				
31	Differential				
32	Overcurrent				
33	Earth-fault				
34	Buchholz				
35	Overtemperature				
36	Pressure-relief				
37	Overvoltage				
38	Field-loss				
39	Surge				
40	Interlocks				
41	Winding temperature				
42	Oil level				
43	Tap position				

3210-MLL-01 SAG MILL: LUBRICATION

Trunnion bearing lubrication oil system data for SAG mill 3210-MLL-01 and drive 3210-MTR-01

The trunnion bearings of SAG mill 3210-MLL-01 are fed by a dedicated oil system with duty and standby pumps, a duplex filter, an oil cooler and a heated reservoir.

Flow, pressure and temperature monitoring is interlocked with drive 3210-MTR-01, 16,500 kW. The mill cannot start, or keep running at 10.13 rpm, without proven lubrication.

No.	Parameter	Unit	Value
1	Mill		3210-MLL-01
2	Drive		3210-MTR-01
3	Power	kW	16,500
4	Speed	rpm	10.13
5	Critical-speed	rpm	13.51
6	Speed-ratio		75.0%
7	Bearings		Trunnions
8	Pumps		Duty/standby
9	Pump-type		Screw
10	Lift-pumps		High-pressure
11	Filter		Duplex
12	Changeover		Online
13	Cooler		Oil/water
14	Coolant		Water
15	Reservoir		Stainless
16	Heater		Yes
17	Oil		EP-mineral
18	Grade		Mill-vendor
19	Flow-switch		Yes
20	Pressure-transmitter		Yes
21	Temperature-element		Yes
22	Level-switch		Yes
23	Differential-pressure		Filter
24	Strainer		Return
25	Trip		Low-pressure
26	Interlock		Startup
27	Rundown		Accumulator
28	Piping		Stainless
29	Control		PLC
30	Breather		Desiccant
31	Sampling		Yes
32	Drain		Valved

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B	2019-10-02	Comments incorporated	Superseded by revision Q			
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NOTES TO SAG DRIVE DATA

These notes record the calculation basis for the SAG gearless drive motor 3210-MTR-01. Rated output is 16,500 kW at 34.5 kV, with power factor 0.95 and efficiency 0.97.

At the site altitude of 3,850 masl, an altitude derating factor of 0.94 applies. This gives a nameplate power of 17,553 kW. Values are rounded as noted.

No.	Note	Applies to
1	Shaft output 16,500 kW.	3210-MTR-01
2	Voltage 34.5 kV.	3210-MTR-01
3	Power factor 0.95.	3210-MTR-01
4	Efficiency 0.97.	3210-MTR-01
5	Full-load current 299.6 A.	3210-MTR-01
6	Input 17,906 kVA.	3210-MTR-01
7	72 poles, 10.13 rpm.	3210-MTR-01
8	Frequency 6.08 Hz.	3210-MTR-01
9	Torque 15,549 kN·m.	3210-MTR-01
10	Angular speed 1.061 rad/s.	3210-MTR-01
11	Altitude 3,850 masl.	Site
12	Derating factor 0.94.	Site
13	Derating applies above 1,000 masl.	Site
14	Unrounded factor 0.94015.	Derating
15	Nameplate 17,553 kW after derating.	3210-MTR-01
16	Design ambient 30 °C.	Site
17	Maximum humidity 95%.	Site
18	Factor rounded to two decimals.	Derating
19	Currents rounded to one decimal.	3210-MTR-01
20	Harmonic limits: Vendor to state	Drive
21		

3210-MLL-01 SAG MILL: INSTRUMENTATION

Mill instrumentation signals wired from 3210-MLL-01 and drive 3210-MTR-01 to local panel 3210-LCP-01.

This page lists the field signals for SAG mill 3210-MLL-01 that are wired to local control panel 3210-LCP-01. They cover speed, power, bearing temperature, vibration, lubrication flow and drive status.

Speed feedback is checked against the operating speed of 10.13 rpm, which is 75.0% of the 13.51 rpm critical speed. Power is read from drive 3210-MTR-01, rated 16,500 kW.

No.	Signal	Unit	Value	Type	Remark
1	Mill tag		3210-MLL-01		
2	Panel		3210-LCP-01		
3	Drive motor		3210-MTR-01		
4	Rated power	kW	16,500		
5	Operating speed	rpm	10.13		
6	Critical speed	rpm	13.51		
7	Speed ratio		75.0%		
8	Speed feedback			Analog	
9	Underspeed alarm				
10	Motor power				
11	Motor current				
12	Bearing temperature				
13	High temperature alarm				
14	High temperature trip				
15	Bearing vibration				
16	Vibration alarm				
17	Vibration trip				
18	Lube flow				
19	Low flow alarm				
20	Low flow trip				
21	Lube pressure				
22	Oil temperature				
23	Pump running				
24	Drive ready				
25	Drive running				
26	Drive fault				
27	Start permissive				
28	Remote start				
29	Emergency stop				
30	Inching mode				
31	Brake applied				
32	Lockout				
33	Shielding				
34	Cabling				
35	Spares				
36	Calibration				
37	Datasheets				
38	Tagging				

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3210-MLL-01 SAG MILL: ALARMS AND TRIPS

SAG mill 3210-MLL-01 and drive alarm and trip list; set points and delays Vendor to state.

This page lists the alarms and trips for SAG mill 3210-MLL-01, which runs at 10.13 rpm, 75.0% of its 13.51 rpm critical speed, and for drive 3210-MTR-01 rated 16,500 kW.

Signals from the cooling-water pump, auxiliary transformer and MCC are grouped by source. Trips stop the mill through 3210-LCP-01; set points are Vendor to state.

No.	Signal	Type/Action	Set point	Unit	Delay	Remark
1	SAG-mill-3210-MLL-01					
2	Mill-speed-high	Trip/stop				10.13 rpm
3	Mill-speed-low	Alarm				
4	Bearing-temp-high	Alarm				
5	Bearing-temp-high-high	Trip/stop				
6	Lube-oil-pressure-low	Alarm				
7	Lube-oil-pressure-low-low	Trip/stop				
8	Lube-oil-flow-low	Trip/stop				
9	Jacking-oil-pressure-low	Permissive				
10	Shell-vibration-high	Alarm				
11	Gear-guard-open	Trip/stop				
12	Inching-mode-active	Alarm				
13	Drive-3210-MTR-01					
14	Motor-overload	Trip/stop				16,500 kW
15	Winding-temp-high	Alarm				
16	Winding-temp-high-high	Trip/stop				
17	Motor-bearing-temp-high	Alarm				
18	Stator-earth-fault	Trip/stop				
19	Air-gap-fault	Trip/stop				
20	Cooling-air-temp-high	Alarm				
21	VFD-fault	Trip/stop				
22	Cooling-pump-3220-MTR-01					
23	Pump-overcurrent	Trip/stop				129.4 A
24	Cooling-water-flow-low	Alarm				
25	Cooling-water-flow-low-low	Trip/stop				
26	Supply-temp-high	Alarm				
27	Transformer-3220-XFR-01					
28	XFR-winding-temp-high	Alarm				
29	XFR-overcurrent	Trip/open				
30	MCC-3220-MCC-01					
31	Incomer-overcurrent	Trip/open				855.7 A
32	Bus-undervoltage	Alarm				
33	Panel-3210-LCP-01					
34	Emergency-stop	Trip/stop				
35	PLC-watchdog-fault	Trip/stop				

3210-MLL-01 SAG MILL: SPARES AND SPECIAL TOOLS

Spares and special tools for SAG mill 3210-MLL-01 and its drive 3210-MTR-01.

This page lists the spare parts and special tools for SAG mill 3210-MLL-01. The mill has an inside diameter of 9.8 m and an effective grinding length of 5.1 m.

The liner handler, torque tools, spare lifters and spare bolts support shell relining. Relining is done with drive 3210-MTR-01, rated 16,500 kW, locked out.

No.	Item	Tag / value	Remark
1	Mill	3210-MLL-01	
2	Diameter	9.8 m	
3	Drive	3210-MTR-01	16,500 kW
4	Special tools		
5	Liner handler		Mill relining
6	Torque wrench		
7	Torque multiplier		
8	Bolt tensioner		
9	Spreader beam		
10	Slings		
11	Shackles		
12	Bolt extractor		
13	Gauges		
14	Spares		
15	Lifter bars		
16	Shell liners		
17	Liner bolts		
18	Nuts		
19	Washers		
20	Seals		
21	Grates		
22	Pulp lifters		
23	Quantities		Vendor to state

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NOTES TO SAG MILL INSTRUMENTATION AND SPARES

These notes cover instrumentation and spares for SAG mill 3210-MLL-01, 9.8 m diameter by 5.1 m effective grinding length, driven by motor 3210-MTR-01 rated 16,500 kW.

Speed monitoring confirms operation at 10.13 rpm, 75.0% of the 13.51 rpm critical speed. Signals from auxiliaries 3220-MTR-01, 3220-XFR-01 and 3220-MCC-01 report to local panel 3210-LCP-01.

No.	Note	Applies to
1	Speed 10.13 rpm	3210-MLL-01
2	Critical 13.51 rpm	3210-MLL-01
3	Overspeed trip: Vendor to state	3210-LCP-01
4	Rated 16,500 kW	3210-MTR-01
5	Winding RTDs	3210-MTR-01
6	Bearing RTDs	3210-MTR-01
7	Vibration probes	3210-MTR-01
8	Current 129.4 A	3220-MTR-01
9	Rated 1,250 kW	3220-MTR-01
10	Impedance 5.0%	3220-XFR-01
11	Winding temperature alarm	3220-XFR-01
12	Load 450 kW	3220-MCC-01
13	Current 855.7 A	3220-MCC-01
14	Hardwired interlocks	3210-LCP-01
15	Local emergency stop	3210-LCP-01
16	Spare instruments: Vendor to state	Instrumentation
17	Commissioning spares included	Spares
18	Operating spares list	Spares
19	Insurance spares: Vendor to state	Spares
20	Spare RTD set	3210-MTR-01
21	Lead time: Vendor to state	Spares

MILL MECHANICAL DATA: 3210-MLL-02 BALL MILL

No.	Parameter	Symbol	Unit	Value	Basis / calculation	Source
1	Mill tag			3210-MLL-02	Tag per plant list	GMDS
2	Mill type			Ball mill	Mill service	SPC
3	Mill inside diameter	D	m	7.9	Inside diameter of shell	DST
4	Effective grinding length	L_eff	m	11.8	Between effective heads	DST
5	Critical speed constant			42.3	Constant in n_c = 42.3 / sqrt(D_eff)	DST
6	Critical speed	n_c	rpm	15.05	42.3 / sqrt(7.9) = 15.0497	DST
7	Operating speed	n_op	rpm	11.29	120 x 6.02 / 64 = 11.2875	DST
8	Operating speed as % of critical		%	75.0%	11.2875 / 15.0497 = 75.00%	DST
9	Mill operating mode			Vendor to state	Process design basis	SPC
10	Drive rated power	P	kW	12,000	Rated mill drive power	SPC
11	Drive pole count	p		64	Gearless motor 3210-MTR-02	DST
12	Drive frequency at operating speed	f	Hz	6.02	Supply frequency	DST
13	Drive angular speed	omega	rad/s	1.182	2 x PI x 11.2875 / 60 = 1.18202	DST
14	Drive torque at operating speed	T	kN·m	10,152	12,000 / 1.18202 = 10,152.1	DST
15	Drive efficiency	eta		0.97	Electrical data basis	DST
16	Design power factor	pf		0.95	Electrical data basis	DST
17	Full-load current at 34.5 kV	I_FL	A	217.9	12,000 / (sqrt(3) x 34.5 x 0.95 x 0.97) = 217.925	DST
18	Drive supply voltage	V	kV	34.5	Medium voltage drive supply	SLD
19	Site altitude		masl	3,850	Site basis	SITE
20	Altitude derating factor			0.94	1 - 0.0000210 x (3,850 - 1,000) = 0.94015, rounded	SPC
21	Drive nameplate power		kW	12,766	12,000 / 0.94 = 12,765.96	DST
22	Input apparent power	S	kVA	13,022	12,000 / (0.95 x 0.97) = 13,022.2	DST
23	Cyclo-converter transformer			3210-XFR-03	Ball drive transformer	SPC
24	Transformer rating		MVA	14.0	Rated power of 3210-XFR-03	DST
25	Transformer impedance		%	7.5%	Impedance of 3210-XFR-03	DST
26	Transformer tap range			±2 x 2.5%	5 positions	DST
27	Mill local control panel			3210-LCP-01	Shared mill operator interface	SPC
28	Shell plate thickness		mm	Vendor to state	Vendor submittal	Vendor
29	Head plate thickness		mm	Vendor to state	Vendor submittal	Vendor
30	Empty mill mass		t	Vendor to state	Vendor submittal	Vendor
31	Charge mass at design load		t	Vendor to state	Vendor submittal	Vendor
32	Trunnion journal diameter		mm	Vendor to state	Vendor submittal	Vendor
33	Trunnion bearing type			Vendor to state	Vendor submittal	Vendor
34	Lubricant flow per bearing		L/min	Vendor to state	Vendor submittal	Vendor
35	Mill rotation direction			Vendor to state	Relative to feed end	Vendor
36	Shell liner type			Vendor to state	Vendor submittal	Vendor
37	Lifter type			Vendor to state	Vendor submittal	Vendor
38	Bearing temperature alarm		°C	Vendor to state	Vendor submittal	Vendor

Values are for revision Q. Entries marked Vendor to state are completed by the mill supplier in its submittal.

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3210-MLL-02 BALL MILL: SHELL AND HEADS

Shell and head data for ball mill 3210-MLL-02 with drive 3210-MTR-02 and site conditions.

The shell and heads of ball mill 3210-MLL-02 form a welded steel drum of 7.9 m inside diameter and 11.8 m effective length. Welds, access openings, flanges and paint are listed below.

The shell carries torque from drive 3210-MTR-02 at 12,000 kW, turning at 11.29 rpm, which is 75.0% of the 15.05 rpm critical speed.

No.	Parameter	Unit	Value	Basis	Source	Status	Remark
1	3210-MLL-02						
2	Diameter	m	7.9				
3	Length	m	11.8				
4	Critical	rpm	15.05				
5	Operating	rpm	11.29				
6	Ratio		75.0%				
7	3210-MTR-02	kW	12,000				
8	Altitude	masl	3,850				
9	Ambient	°C	30				
10	Minimum	°C	-12				
11	Maximum	°C	27				
12	Wind	m/s	36				
13	Zone		4				
14	PGA	g	0.42				
15	Class		D				
16	Frost	mm	450				
17	Humidity		58%				
18	Humidity max		95%				
19	Derating		0.94				
20	Shell						
21	Material		Steel				
22	Joints		Welded				
23	Certificate		Yes				
24	Heads						
25	Material		Steel				
26	Form		Fabricated				
27	Welds						
28	Type		Butt				
29	Radiography		Yes				
30	Ultrasonic		Yes				
31	Stress-relief		Yes				
32	Access						
33	Manholes		Yes				
34	Covers		Bolted				
35	Flanges						
36	Faces		Machined				
37	Bolting		Fitted				
38	Painting						
39	Exterior		Epoxy				
40	Interior		Bare				
41	Mass						
42	Certificate		Yes				
43	Balance		Static				
44	Lifting		Lugs				
45	Inspection		Witnessed				

3210-MLL-02 BALL MILL: TRUNNIONS AND BEARINGS

Trunnion, bearing, seal and pedestal data for ball mill 3210-MLL-02, including mill, drive and site values.

The 3210-MLL-02 ball mill has a 7.9 m inside diameter and an 11.8 m effective length. It rests on trunnion journals carried by hydrostatic bearings in fabricated housings on cast pedestals, with labyrinth seals.

The bearings carry the shell at 11.29 rpm, which is 75.0% of critical, under drive 3210-MTR-02 rated 12,000 kW. Bearing RTDs and vibration sensors send alarm and trip signals.

No.	Parameter	Unit	Value	Basis	Source	Remark
1	Mill		3210-MLL-02			
2	Drive		3210-MTR-02			
3	Power	kW	12,000			
4	Diameter	m	7.9			
5	Length	m	11.8			
6	Critical	rpm	15.05			
7	Operating	rpm	11.29			
8	Ratio		75.0%			
9	Altitude	masl	3,850			
10	Derating		0.94			
11	Ambient	°C	30			
12	Minimum	°C	-12			
13	Maximum	°C	27			
14	Humidity		95%			
15	Seismic		4			
16	PGA	g	0.42			
17	Class		D			
18	Journals		Forged			
19	Finish		Polished			
20	Bearings		Hydrostatic			
21	Liners		Babbitt			
22	Shoes		Spherical			
23	Housings		Fabricated			
24	Split		Horizontal			
25	Seals		Labyrinth			
26	Purge		Yes			
27	Pedestals		Cast			
28	Grout		Epoxy			
29	Foundation		Concrete			
30	Expansion		Free-end			
31	Locating		Feed-end			
32	Lubrication		HP/LP			
33	Lift-pumps		Yes			
34	Accumulator		Yes			
35	RTDs		Yes			
36	Temp-alarm		Yes			
37	Temp-trip		Yes			
38	Vibration		Monitored			
39	Accelerometers		Yes			
40	Oil-pressure		Yes			
41	Interlocks		DCS			
42	Signals		Hardwired			
43	Thermowells		Yes			
44	Covers		Removable			
45	Jacking		Provided			
46	Isolation		Insulated			
47	Paint		Epoxy			
48	Heaters		Yes			
49	Oil-cooler		Yes			
50	Filters		Duplex			

3210-MLL-02 BALL MILL: FEED AND DISCHARGE

Feed and discharge arrangement data for ball mill 3210-MLL-02, including chute, trunnions, grate, pulp lifters and trommel.

Slurry enters ball mill 3210-MLL-02 through a lined feed chute and the feed trunnion. With an inside diameter of 7.9 m and effective length of 11.8 m, the trunnion bores must pass the full circulating load.

At discharge, a grate and pulp lifters lift slurry to the outlet trunnion and trommel. The mill runs at 11.29 rpm, 75.0% of critical, driven by 3210-MTR-02.

No.	Parameter	Unit	Value	Basis	Remark
1	Mill		3210-MLL-02		
2	Diameter	m	7.9		
3	Effective length	m	11.8		
4	Critical speed	rpm	15.05		
5	Speed	rpm	11.29		
6	Speed ratio		75.0%		
7	Drive		3210-MTR-02		
8	Power	kW	12,000		
9	Feeding		Chute		
10	Scoop		No		
11	Lining		Rubber		
12	Spout		Cantilevered		
13	Seal		Labyrinth		
14	Inlet		Trunnion		
15	Liner		Replaceable		
16	Bellmouth		Yes		
17	Flushing		Yes		
18	Access		Hatch		
19	Discharge		Grate		
20	Grate		Segmented		
21	Material		Steel		
22	Slots		Tapered		
23	Orientation		Radial		
24	Lifters		Pulp		
25	Profile		Curved		
26	Cone		Central		
27	Fastening		Bolted		
28	Outlet		Trunnion		
29	Insert		Replaceable		
30	Spiral		Reversing		
31	Trommel		Yes		
32	Mounting		Trunnion		
33	Panels		Polyurethane		
34	Apertures		Slotted		
35	Spray		Yes		
36	Rejects		Chute		
37	Undersize		Hopper		
38	Hood		Enclosed		
39	Drain		Yes		
40	Inspection		Visual		
41	Handling		Hoist		
42	Guarding		Yes		
43	Dust		Contained		
44	Painting		Epoxy		
45	Bolts		Through		
46	Gaskets		Rubber		
47	Spares		Listed		
48	Drawings		Vendor		
49	Lifting		Lugs		
50	Relining		Machine		

3210-MLL-02 BALL MILL: SHELL LINERS

Shell liner arrangement, material, fixing, handling and spares for ball mill 3210-MLL-02.

Shell liners protect the shell of ball mill 3210-MLL-02, which has a 7.9 m inside diameter and 11.8 m effective length. Lifter and wave rows set the charge trajectory at 11.29 rpm, or 75.0% of the 15.05 rpm critical speed.

Liner wear and profile govern power draw from drive 3210-MTR-02, rated 12,000 kW. Relining uses a handler and inching drive.

No.	Item description	Unit	Value	Basis	Remark
1	Mill		3210-MLL-02		
2	Drive		3210-MTR-02		
3	Power	kW	12,000		
4	Diameter	m	7.9		
5	Length	m	11.8		
6	Critical	rpm	15.05	42.3/√D_eff	
7	Operating	rpm	11.29		
8	Ratio		75.0%	Of critical	
9	Lining		Required		
10	Arrangement		Row-wise		
11	Feed-row		Lifter		
12	Mid-rows		Wave		
13	Discharge-row		Lifter		
14	Profile		Asymmetric		
15	Orientation		Directional		
16	Rotation		Single		
17	Material		Cr-Mo		
18	Alternative		Rubber-steel		
19	Casting		Heat-treated		
20	Testing		Certified		
21	Fixing		Bolted		
22	Bolts		Oval-head		
23	Nuts		Locking		
24	Washers		Sealing		
25	Backing		Rubber		
26	Shell-drilling		Shop		
27	Handling		Handler		
28	Handler-drive		Hydraulic		
29	Access		Feed-end		
30	Inching		Required		
31	Inching-drive		Auxiliary		
32	Lifting-lugs		Cast-in		
33	Tooling		Supplied		
34	Spares		Set		
35	Spare-bolts		Set		
36	Storage		Covered		
37	Wear-monitoring		Ultrasonic		
38	Profiling		Laser		
39	Replacement		Row-wise		
40	Discard-criterion		Profile		
41	Liner-drawings		Required		
42	Torque-sheet		Required		
43	Weights-list		Required		
44	Installation		Supervised		
45	Commissioning		Inspection		
46	Warranty		Required		
47	Interchangeability		Required		
48	Marking		Cast		
49	Packing		Crated		
50	Certification		Material		

3210-MLL-02 BALL MILL: LIFTER SCHEDULE

Lifter schedule for ball mill 3210-MLL-02, covering mill data, lifter design, fixing and replacement set.

Lifters line the shell of ball mill 3210-MLL-02, which has an inside diameter of 7.9 m and an effective length of 11.8 m. Their profile and row layout set how the charge is lifted and released.

The mill runs at 11.29 rpm, 75.0% of the 15.05 rpm critical speed, driven by 3210-MTR-02 at 12,000 kW. Lifter dimensions and row count: Vendor to state.

No.	Parameter	Unit	Value	Detail	Remark
1	Mill				
2	Tag		3210-MLL-02		
3	ID	m	7.9		
4	Eff. length	m	11.8		
5	Critical speed	rpm	15.05		
6	Speed	rpm	11.29		
7	Ratio		75.0%		
8	Drive		3210-MTR-02		
9	Power	kW	12,000		
10	Lifters				
11	Type		Wave		
12	Profile		Trapezoidal		
13	Base		Flat		
14	Material		Cr-Mo steel		
15	Hardened		Yes		
16	Layout		Uniform		
17	Fixing				
18	Bolting		Through-shell		
19	Head		Oval		
20	Seal		Washer		
21	Nut		Locking		
22	Spares				
23	Replacement		Full set		
24	Bolts		Included		
25	Seals		Included		
26	Handler		Required		

3210-MLL-02 BALL MILL: HEAD AND DISCHARGE LINERS

Head, grate, pulp lifter and sealing liner data for ball mill 3210-MLL-02

This page covers the feed and discharge head liners of ball mill 3210-MLL-02, including the grate, pulp lifter liners and the sealing between liners and heads.

The liners must suit a 7.9 m inside diameter, 11.8 m effective length and an operating speed of 11.29 rpm, 75.0% of critical, with drive 3210-MTR-02 rated 12,000 kW.

No.	Parameter	Unit	Value	Basis	Remark
1	3210-MLL-02				
2	Diameter	m	7.9		
3	Speed	rpm	11.29		
4	Drive	kW	12,000		
5	Feed				
6	Material		Rubber		
7	Thickness: Vendor to state				
8	Fixing		Bolted		
9	Discharge				
10	Material		Rubber		
11	Thickness: Vendor to state				
12	Grate				
13	Material		Steel		
14	Slot: Vendor to state				
15	Lifters				
16	Material		Steel		
17	Count: Vendor to state				
18	Sealing				
19	Gaskets				
20	Sealed bolts				
21	Trunnion seals				

3210-MLL-02 BALL MILL: LINER WEAR AND REPLACEMENT

Ball mill 3210-MLL-02 liner wear points, replacement limits, relining steps and records; limits set by vendor.

Liner wear on ball mill 3210-MLL-02 is tracked at fixed points on the feed head, shell rows and discharge head. The 7.9 m diameter and 11.8 m effective length set the survey grid.

At 11.29 rpm, 75.0% of the 15.05 rpm critical speed, lifter wear changes charge action. Replacement limits are vendor stated; relining uses drive 3210-MTR-02 inching.

No.	Item	Unit	Value	Remark
1	Tag		3210-MLL-02	
2	Diameter	m	7.9	
3	Length	m	11.8	
4	Critical speed	rpm	15.05	
5	Operating speed	rpm	11.29	75.0%
6	Drive		3210-MTR-02	
7	Power	kW	12,000	
8	Lifter limit	mm	Vendor to state	
9	Plate limit	mm	Vendor to state	
10	Interval	h	Vendor to state	
11	Measurement points			
12	Feed head			
13	Shell rows			
14	Discharge head			
15	Laser scan			
16	Isolation			
17	Inching			
18	Drawdown			
19	Removal			
20	Inspection			
21	Installation			
22	Bolt retorque			
23	Restart			
24	Logbook			
25	Photographs			
26	Thickness			
27	Trend			
28	Sign-off			
29	Handover			
30	Archive			

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NOTES TO BALL MILL MECHANICAL DATA

These notes set the basis for the mechanical data of ball mill 3210-MLL-02. An inside diameter of 7.9 m and an effective length of 11.8 m give a critical speed of 15.05 rpm. The operating speed of 11.29 rpm is 75.0% of critical.

Drive 3210-MTR-02 is rated 12,000 kW and is derated by a factor of 0.94 for 3,850 masl. The vendor confirms all values.

No.	Note	Applies to
1	Design basis values	All
2	Vendor verifies values	All
3	Inside diameter 7.9 m	3210-MLL-02
4	Effective length 11.8 m	3210-MLL-02
5	Critical speed 15.05 rpm	3210-MLL-02
6	n_c = 42.3/sqrt(D_eff)	3210-MLL-02
7	Operating 11.29 rpm, 75.0%	3210-MLL-02
8	Drive rated 12,000 kW	3210-MTR-02
9	Altitude 3,850 masl	Site
10	Derating factor 0.94	3210-MTR-02
11	Design ambient 30 °C	Site
12	Extreme minimum -12 °C	Site
13	Extreme maximum 27 °C	Site
14	Design wind 36 m/s	Site
15	Zone 4, PGA 0.42 g	Foundations
16	Site class D	Foundations
17	Frost depth 450 mm	Foundations
18	Humidity 58% mean, 95% max	Site
19	Mass: Vendor to state	
20	Drawings govern	
21	Vendor responsible	

3210-MLL-02 BALL MILL: DRIVE MOTOR INTERFACE

Ball mill 3210-MLL-02 gearless motor rotor mounting, air gap, bolting, alignment and access data

The rotor of drive 3210-MTR-02 is mounted directly on the shell of ball mill 3210-MLL-02, which has a 7.9 m inside diameter and 11.8 m effective length. The shell carries the rotor and transmits 10,152 kN·m torque.

At 11.29 rpm, 75.0% of critical speed, the air gap, bolting and alignment must stay stable. Access covers allow inspection.

No.	Parameter	Unit	Value	Remark
1	Mill		3210-MLL-02	
2	Motor		3210-MTR-02	
3	Diameter	m	7.9	
4	Length	m	11.8	
5	Speed	rpm	11.29	
6	Critical	rpm	15.05	
7	Ratio		75.0%	
8	Power	kW	12,000	
9	Torque	kN·m	10,152	
10	Poles		64	
11	Frequency	Hz	6.02	
12	Gap	mm	Vendor to state	
13	Rotor			
14	Mounting		Flange	
15	Segments		Bolted	
16	Support		Shell	
17	Stiffness		Verified	
18	Material		Steel	
19	Bolting			
20	Bolts		Prestressed	
21	Tightening		Hydraulic	
22	Locking		Required	
23	Inspection		Periodic	
24	Alignment			
25	Method		Laser	
26	Centring		Concentric	
27	Runout		Monitored	
28	Sensors		Fitted	
29	Shims		Stator	
30	Access			
31	Platforms		Provided	
32	Covers		Removable	
33	Doors		Interlocked	
34	Lighting		Yes	
35	Lifting		Provided	
36	Brakes		Yes	
37	Inching		Yes	
38	Lockout		Yes	

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3210-MTR-02 BALL DRIVE: ELECTRICAL DATA

Electrical ratings of gearless drive motor 3210-MTR-02 and its supply for ball mill 3210-MLL-02.

Gearless motor 3210-MTR-02 drives ball mill 3210-MLL-02 at 11.29 rpm, which is 75.0% of the 15.05 rpm critical speed. Its 64 poles run at 6.02 Hz from a 34.5 kV supply, giving 10,152 kN·m torque.

At 3,850 masl a derating factor of 0.94 applies, so the 12,000 kW shaft rating needs a 12,766 kW nameplate and 13,022 kVA input.

No.	Parameter	Unit	Value	Basis	Source	Remark
1	Identification					
2	Drive		3210-MTR-02			
3	Mill		3210-MLL-02			
4	Type		Gearless			
5	Motor					
6	Power	kW	12,000			
7	Nameplate	kW	12,766			
8	Efficiency		0.97			
9	Poles		64			
10	Torque	kN·m	10,152			
11	Supply					
12	Voltage	kV	34.5			
13	Current	A	217.9			
14	PF		0.95			
15	Apparent	kVA	13,022			
16	Frequency	Hz	6.02			
17	Speed					
18	Motor	rpm	11.29			
19	Angular	rad/s	1.182			
20	Derating					
21	Factor		0.94			
22	Unrounded		0.94015			
23	Coefficient		0.0000210			
24	Reference	masl	1,000			
25	Site					
26	Altitude	masl	3,850			
27	Ambient	°C	30			
28	Minimum	°C	-12			
29	Maximum	°C	27			
30	Mean RH		58%			
31	Max RH		95%			
32	Mill					
33	Diameter	m	7.9			
34	Length	m	11.8			
35	Critical	rpm	15.05			
36	Operating	rpm	11.29			
37	Fraction		75.0%			
38	Auxiliaries					
39	Cooling		Forced			
40	Heaters		Yes			
41	Converter		Static			
42	Inching		Yes			
43	Brake		Yes			
44	Protection					
45	Earthing		Required			
46	Relays		Vendor			
47	RTDs		Required			
48	Vibration		Required			
49	Notes					
50	Values		Fictional			

3210-MTR-02 BALL DRIVE: TRANSFORMERS AND EXCITATION

Ball drive cyclo-converter and excitation transformers, rectifier, protection, system voltages and harmonic limits for 3210-MTR-02.

Transformer 3210-XFR-03, rated 14.0 MVA with 7.5% impedance, feeds the cyclo-converter of the 12,000 kW ball mill motor 3210-MTR-02 from the 34.5 kV drive supply. The motor draws 13,022 kVA, within the transformer rating.

Excitation transformer 3210-XFR-01, 1.6 MVA at 34.5 kV / 380 V, supplies the field rectifier. Filter 3110-FLT-01 keeps harmonics within IEEE 519-2014 limits.

No.	Parameter	Unit	Value	Basis	Tag	Remark
1	3210-XFR-03					
2	Rating	MVA	14.0			
3	Impedance		7.5%			
4	Primary	kV	34.5			
5	Secondary: Vendor to state					
6	Taps		±2x2.5%			
7	Positions		5			
8	Cooling		Oil-filled			
9	Windings		Copper			
10	3210-XFR-01					
11	Rating	MVA	1.6			
12	Primary	kV	34.5			
13	Secondary	V	380			
14	Impedance		6.0%			
15	Type		Dry			
16	Duty		Excitation			
17	Rectifier					
18	Supply	V	380			
19	Type		Thyristor			
20	Cooling		Forced-air			
21	Output		DC			
22	Protection					
23	Differential		Yes			
24	Overcurrent		Yes			
25	Earth-fault		Yes			
26	Buchholz		Yes			
27	Temperature		Yes			
28	Monitoring					
29	Oil-level		Yes			
30	System					
31	MV	kV	23/34.5			
32	Drive	kV	34.5			
33	Auxiliary	kV	6.6			
34	LV	V	380			
35	PCC-PF		0.95	Lagging		
36	IEEE 519-2014					
37	Voltage THD		5.0%			
38	Individual		3.0%			
39	TDD		5.0%			
40	3110-FLT-01					
41	Rating	Mvar	6.0			
42	Voltage	kV	34.5			
43	Tuned		5th			
44	3210-MTR-02					
45	Input	kVA	13,022			
46	Efficiency		0.97			
47	PF		0.95			

AUXILIARY SUPPLIES TO THE MILL DRIVES

Auxiliary motor, transformer and motor control center supplies serving the mill drives at 34.5 kV

Motor 3220-MTR-01 drives the cooling-water pump at 1,250 kW and 6.6 kV from the auxiliary bus. Transformer 3220-XFR-01, rated 0.8 MVA, steps 6.6 kV down to 380 V for 3220-MCC-01, which carries 450 kW.

These loads support the 34.5 kV mill drives and panel 3210-LCP-01. Filter 3110-FLT-01 holds harmonics within IEEE 519-2014 limits.

No.	Parameter	kV	kW	A	pf / %	Remark
1	3220-MTR-01					
2	Rating		1,250			
3	Voltage	6.6				
4	pf				0.88	
5	Efficiency				0.96	
6	Current			129.4		
7	3220-XFR-01					
8	Rating					0.8 MVA
9	Primary	6.6				
10	Secondary					380 V
11	Impedance				5.0%	
12	3220-MCC-01					
13	Voltage					380 V
14	Load		450			
15	pf				0.85	
16	Efficiency				0.94	
17	Current			855.7		
18	3210-LCP-01					
19	MV	23/34.5				
20	Drive	34.5				
21	Auxiliary	6.6				
22	LV					380 V
23	pf				≥0.95	Lagging
24	THD				5.0%	
25	Individual				3.0%	
26	TDD				5.0%	
27	3110-FLT-01	34.5				6.0 Mvar

3210-MLL-02 BALL MILL: LUBRICATION

Trunnion bearing lubrication oil system data for ball mill 3210-MLL-02 driven by 3210-MTR-02.

The trunnion bearings of ball mill 3210-MLL-02 are served by a dedicated oil system with duty and standby pumps, a duplex filter, an oil cooler and a heated reservoir.

Oil flow, pressure and temperature are monitored and form start permissives for drive 3210-MTR-02, rated 12,000 kW, so the mill cannot turn at 11.29 rpm without proven lubrication.

No.	Parameter	Unit	Value
1	Mill		3210-MLL-02
2	Drive		3210-MTR-02
3	Power	kW	12,000
4	Speed	rpm	11.29
5	PUMPS		
6	Arrangement		Duty/standby
7	Type		Gear
8	Flow	L/min	Vendor to state
9	FILTER		
10	Type		Duplex
11	Alarm		Differential
12	COOLER		
13	Type		Oil/water
14	Duty	kW	Vendor to state
15	RESERVOIR		
16	Material		Stainless
17	Volume	L	Vendor to state
18	Heater		Yes
19	MONITORING		
20	Flow		Switch
21	Pressure		Transmitter
22	Temperature		Transmitter
23	Level		Switch
24	Interlock		Permissive
25	OIL		
26	Grade		Vendor to state
27	Type		Mineral
28	BEARINGS		
29	Lift		Hydrostatic
30	Lubrication		Hydrodynamic
31	Feed		Dual
32	NOTES		
33	Drawings		Vendor

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NOTES TO BALL DRIVE DATA

These notes record the calculation basis for the ball gearless drive motor 3210-MTR-02. The 12,000 kW rating is supplied at 34.5 kV, with power factor 0.95 and efficiency 0.97.

At a site altitude of 3,850 masl, an altitude derating factor of 0.94 applies. This raises the nameplate power to 12,766 kW. Values are rounded as listed.

No.	Note	Applies to
1	Rated output 12,000 kW	3210-MTR-02
2	Supply 34.5 kV	3210-MTR-02
3	Power factor 0.95	3210-MTR-02
4	Efficiency 0.97	3210-MTR-02
5	Current 217.9 A	3210-MTR-02
6	Input 13,022 kVA	3210-MTR-02
7	Nameplate 12,766 kW	3210-MTR-02
8	Torque 10,152 kN·m	3210-MTR-02
9	11.29 rpm, 1.182 rad/s	3210-MTR-02
10	64 poles, 6.02 Hz	3210-MTR-02
11	Altitude derating factor 0.94	3210-MTR-02
12	Site altitude 3,850 masl	Site
13	0.0000210 per m above 1,000 masl	
14	0.94015 rounded to 0.94	
15	Nameplate equals rating divided by 0.94	
16	Design ambient 30 °C	
17	Extremes -12 °C and 27 °C	
18	Humidity 58% mean, 95% max	
19	Harmonics: Vendor to state	
20	Rounded values shown	
21	Inrush: Vendor to state	

3210-MLL-02 BALL MILL: INSTRUMENTATION

Ball mill 3210-MLL-02 instrument signals wired to local control panel 3210-LCP-01, with key mill and drive values.

Speed, power, bearing temperature, vibration, lubrication flow and drive status signals from mill 3210-MLL-02 are wired to local panel 3210-LCP-01 for monitoring, alarms and interlocks.

The speed signal confirms operation at 11.29 rpm, which is 75.0% of the 15.05 rpm critical speed for the 7.9 m mill. Power is monitored against the 12,000 kW rating of drive 3210-MTR-02.

No.	Signal	I/O	Unit	Value	Remark
1	Speed	AI	rpm	11.29	
2	Critical		rpm	15.05	
3	Ratio			75.0%	
4	Power	AI	kW	12,000	
5	Drive			3210-MTR-02	
6	Current	AI			
7	Stator-temp	RTD			
8	Motor-bearing-temp	RTD			
9	Inlet-trunnion-temp	RTD			
10	Outlet-trunnion-temp	RTD			
11	Pinion-bearing-temp	RTD			
12	Motor-vibration	AI			
13	Pinion-vibration	AI			
14	Gearbox-vibration	AI			
15	Lube-flow	AI			
16	Lube-pressure	AI			
17	Lube-temperature	RTD			
18	Oil-level	DI			
19	Filter-dP	DI			
20	Lube-pump-run	DI			
21	Jacking-pump-run	DI			
22	Drive-ready	DI			
23	Drive-running	DI			
24	Drive-trip	DI			
25	Start-command	DO			
26	Stop-command	DO			
27	Emergency-stop	DI			
28	Inching-mode	DI			
29	Clutch-engaged	DI			
30	Brake-applied	DI			
31	Zero-speed	DI			
32	Gear-spray	DI			
33	Remote/local	DI			
34	Fault-reset	DO			
35	Diameter		m	7.9	
36	Length		m	11.8	
37	Panel			3210-LCP-01	
38	Interlock-healthy	DI			

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3210-MLL-02 BALL MILL: ALARMS AND TRIPS

Alarm and trip list for ball mill 3210-MLL-02 and its drive; set points Vendor to state.

This page lists the alarms and trips for ball mill 3210-MLL-02 and its 12,000 kW drive 3210-MTR-02. The mill has a 7.9 m inside diameter and an 11.8 m effective length. Signals are grouped by system and wired to local control panel 3210-LCP-01.

Speed deviation is checked against the 11.29 rpm operating speed, which is 75.0% of critical. All set points are Vendor to state.

No.	Signal	Type	Source	Action	Set point	Remark
1	Drive-motor		3210-MTR-02			
2	Stator-winding-temp-high	Alarm				
3	Stator-winding-temp-HH	Trip		Stop-mill		
4	Motor-bearing-temp-high	Alarm				
5	Motor-bearing-temp-HH	Trip		Stop-mill		
6	Motor-overcurrent	Trip		Stop-mill		
7	Motor-vibration-high	Alarm				
8	Mill-bearings		3210-LCP-01			
9	Trunnion-bearing-temp-high	Alarm				
10	Trunnion-bearing-temp-HH	Trip		Stop-mill		
11	Bearing-flow-low	Alarm				
12	Mill-vibration-high	Alarm				
13	Lubrication		3210-LCP-01			
14	Lube-oil-pressure-low	Alarm				
15	Lube-oil-pressure-LL	Trip		Stop-mill		
16	Lube-oil-temp-high	Alarm				
17	Lube-tank-level-low	Alarm				
18	Lube-filter-dP-high	Alarm				
19	Jacking-oil-pressure-low	Trip				
20	Cooling-water		3220-MTR-01			
21	Pump-motor-trip	Alarm				
22	CW-flow-low	Alarm				
23	CW-supply-temp-high	Alarm				
24	CW-flow-LL	Trip		Stop-mill		
25	Gear-and-pinion		3210-LCP-01			
26	Pinion-bearing-temp-high	Alarm				
27	Gear-spray-fault	Alarm				
28	Inching-drive-engaged	Trip				
29	Girth-gear-guard-open	Trip				
30	Electrical		3220-MCC-01			
31	MCC-supply-fault	Alarm				
32	Aux-transformer-temp-high	Alarm				
33	Speed-deviation	Alarm				
34	Emergency-stop	Trip				
35	LCP-power-fail	Alarm				

SPARES, SPECIAL TOOLS AND DOCUMENTATION

Spares, special tools, documentation and applicable standards for the SAG mill and ball mill packages

This page lists spares, special tools and documents to be supplied with SAG mill 3210-MLL-01 and ball mill 3210-MLL-02. Spares cover liners, bearings, seals and the drive packages 3210-MTR-01 and 3210-MTR-02.

Drive documents from Veltrane Drives GmbH shall cover the 16,500 kW and 12,000 kW motors. Work shall follow the listed standards and LAHJ requirements.

No.	Item	Applies to	Requirement
1	Spares		
2	Liner bolts	Both	
3	Shell liners	3210-MLL-01	
4	Shell liners	3210-MLL-02	
5	Grates	3210-MLL-01	
6	Bearing pads	Both	
7	Trunnion seals	Both	
8	Motor spares	3210-MTR-01	
9	Motor spares	3210-MTR-02	
10	Drive cards	Veltrane	
11	Quantities		Vendor to state
12	Special tools		
13	Liner handler	Both	
14	Inching drive	Both	
15	Jacks		
16	Torque wrenches	Both	
17	Documentation		
18	Drawings	Both	
19	Manuals	Both	
20	Test reports	Drives	
21	Databooks	Both	
22	Rating	3210-MTR-01	16,500 kW
23	Rating	3210-MTR-02	12,000 kW
24	Standards		
25	IEC 60034-1		
26	IEEE 519-2014		
27	ISO 9001:2015		
28	NEMA MG 1		
29	ANSI C57.12.00		
30	NAEC-2019		
31	LAHJ		
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NOTES

These closing notes set the units, calculation basis and references used across the data sheet. Entries marked Vendor to state are completed by the supplier.

Mill speeds use n_c = 42.3 / sqrt(D_eff), giving 75.0% of critical for both mills. Drive ratings of 16,500 kW and 12,000 kW are subject to the 0.94 altitude derating factor.

No.	Note	Applies to
1	SI units	All
2	Power in kW	Drives
3	n_c = 42.3 / sqrt(D_eff)	Mills
4	Speed 75.0% of critical	Mills
5	Derating factor 0.94	Motors
6	Altitude 3,850 masl	Site
7	Design ambient 30 °C	Motors
8	Blanks: Vendor to state	Vendor
9	Drive vendor Veltrane Drives GmbH	Drives
10	IEC 60034-1 applies	Motors
11	NEMA MG 1 applies	Motors
12	IEEE 519-2014 harmonics	Drives
13	ANSI C57.12.00 applies	Transformers
14	NAEC-2019 installation	Electrical
15	LAHJ approval required	Site
16	ISO 9001:2015 quality	Vendor
17	Seismic zone 4	Structures
18	Client Bramhall Copper Corp.	Project
19	Contractor: Ostrander Engineering Services Ltd.	Project
20	Quebrada Azul Copper Expansion	Project
21	Revision: Q	All

End of data sheet.